# ERGA Assembly Report

v24.10.15

Tags: ERGA-BGE

| | |
|---|---|
| TxID | 3230685 |
| ToLID | **icPseApti1** |
| Species | Pseudanchomenus aptinoides |
| Class | Insecta |
| Order | Coleoptera |

| Genome Traits | Expected | Observed |
|---|---|---|
| Haploid size (bp) | 828,164,504 | 1,011,527,272 |
| Haploid Number | 19 (source: ancestor) | 21 |
| Ploidy | 2 (source: ancestor) | 2 |
| Sample Sex | Unknown | Unknown |

## EBP metrics summary and curation notes

Obtained EBP quality metric for collapsed: 6.7.Q62

The following metrics were automatically flagged as below EBP recommended standards or different from expected:

. Observed Haploid size (bp) has >20% difference with Expected
. Observed Haploid Number is different from Expected

. Kmer completeness value is less than 90 for collapsed
. BUSCO single copy value is less than 90% for collapsed
. Assembly length loss > 3% for collapsed
. Not 90% of assembly in chromosomes for collapsed

### Curator notes

. Interventions/Gb: 274
. Contamination notes: ""
. Other observations: "The assembly of Pseudanchomenus aptinoides (icPseApti1) is based on 33X PacBio data and 181X Arima Hi-C data generated as part of the European Reference Genome Atlas (ERGA, https://www.erga-biodiversity.eu/) via the Biodiversity Genomics Europe project (BGE, https://biodiversitygenomics.eu/). The assembly process included the following steps: initial PacBio assembly generation with Hifiasm, removal of contaminant sequences using Context, removal of haplotypic duplications using purge_dups, and Hi-C-based scaffolding with YaHS. In total, 35 contigs were identified as contaminants (bacterial, archaeal, or viral), totaling 9.446 Mb (with the largest being 4.384 Mb). Additionally, 1145 regions totaling 121.128 Mb (with the largest being 0.567 Mb) were identified as haplotypic duplications and removed. The mitochondrial genome was assembled using OATK. Finally, the primary assembly was analyzed and manually improved using Pretext. During manual curation, 11 haplotypic

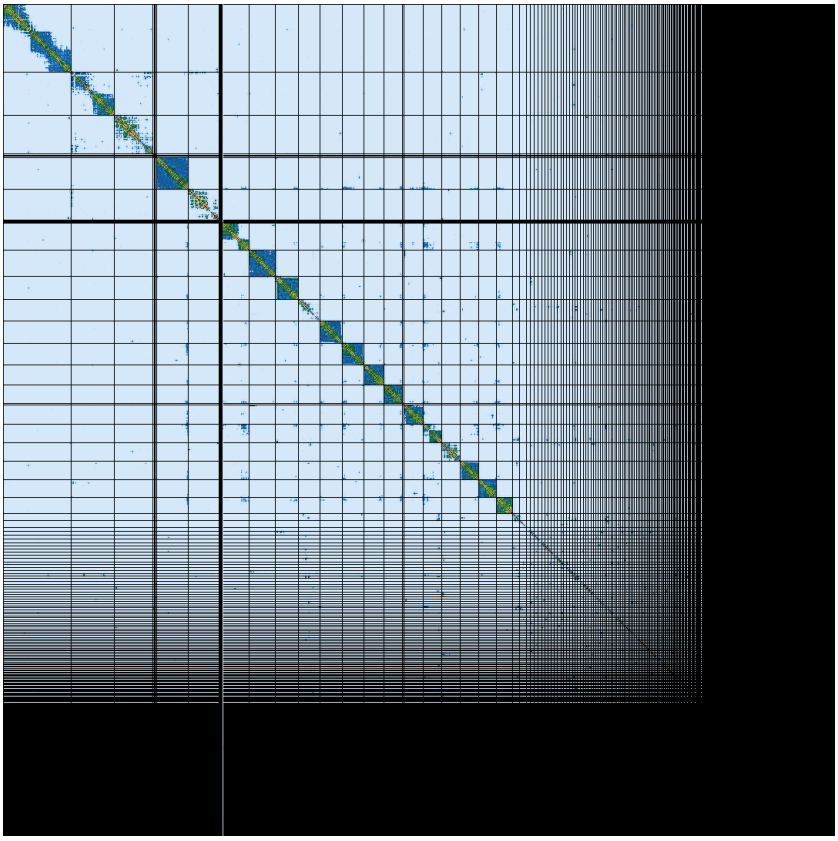
regions were removed, totaling 8.298Mb , (with the largest being 1.993Mb).Chromosome-scale scaffolds confirmed by Hi-C data were named in order of size. "

# Quality metrics table

| Metrics | Pre-curation collapsed | Curated collapsed |
|---|---|---|
| Total bp | 1,089,488,111 | 1,011,527,272 |
| GC % | 31.53 | 31.53 |
| Gaps/Gbp | 405.7 | 409.28 |
| Total gap bp | 45,200 | 53,500 |
| Scaffolds | 1,079 | 652 |
| Scaffold N50 | 6,363,035 | 22,348,468 |
| Scaffold L50 | 27 | 15 |
| Scaffold L90 | 375 | 203 |
| Contigs | 1,504 | 1,066 |
| Contig N50 | 2,362,129 | 2,847,141 |
| Contig L50 | 69 | 58 |
| Contig L90 | 629 | 474 |
| QV | 62.3893 | 62.3073 |
| Kmer compl. | 85.4298 | 78.307 |
| BUSCO sing. | 90.5% | 84.8% |
| BUSCO dupl. | 5.1% | 4.7% |
| BUSCO frag. | 1.5% | 0.5% |
| BUSCO miss. | 2.8% | 10.1% |

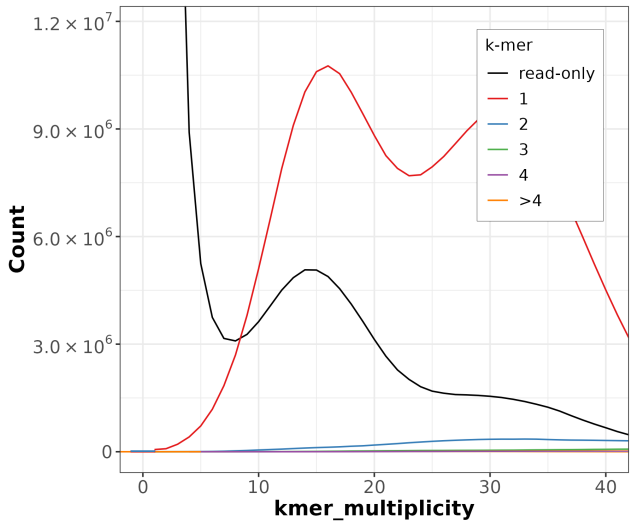
Warning! BUSCO versions or lineage datasets are not the same across results:
BUSCO: 5.8.2 (euk_genome_met, metaeuk) / Lineage: coleoptera_odb12 (genomes:64, BUSCOs:3729)
BUSCO: 6.0.0 (euk_genome_min, miniprot) / Lineage: coleoptera_odb12 (genomes:64, BUSCOs:3729)

# HiC contact map of curated assembly

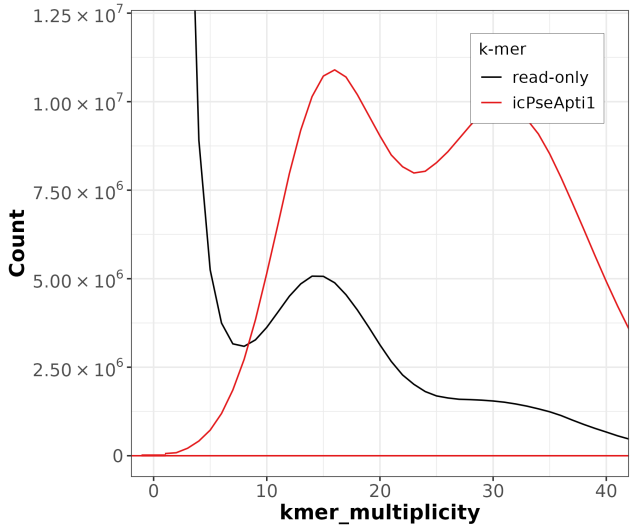


**collapsed** [LINK]

# K-mer spectra of curated assembly

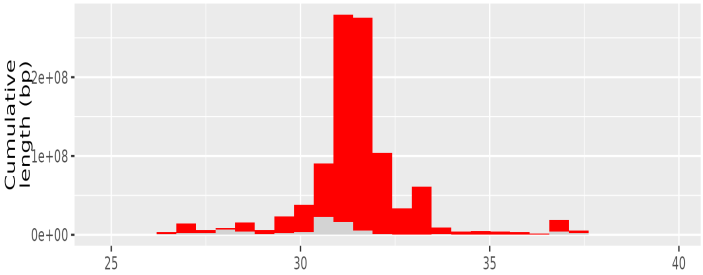



Distribution of k-mer counts per copy numbers found in asm

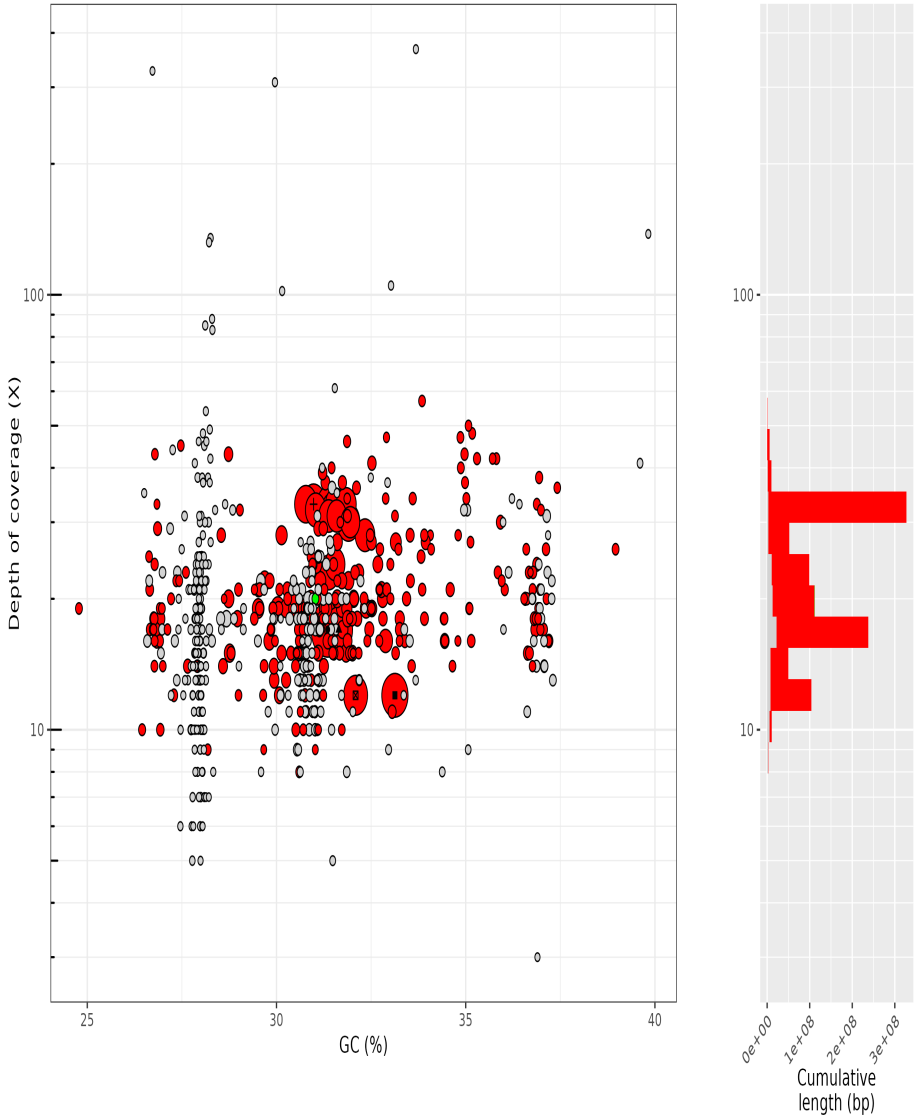



Distribution of k-mer counts coloured by their presence in reads/assemblies

# Post-curation contamination screening




TAPAs summary Graph




**collapsed.** Bubble plot circles are scaled by sequence length, positioned by coverage and GC proportion, and coloured by taxonomy. Histograms show total assembly length distribution on each axis.

# Data profile

| Data | PACBIO Hifi | Arima |
|---|---|---|
| Coverage | 33 | 181 |

# Assembly pipeline

- **Hifiasm**
    |_ *ver:* 0.19.5-r593
    |_ *key param:* NA
- **purge_dups**
    |_ *ver:* 1.2.5
    |_ *key param:* NA
- **YaHS**
    |_ *ver:* 1.2
    |_ *key param:* NA

# Curation pipeline

- **PretextMap**
    |_ *ver:* 0.1.9
    |_ *key param:* NA
- **PretextView**
    |_ *ver:* 0.2.5
    |_ *key param:* NA

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Date and time: 2026-03-13 08:16:10 CET